which the waves' oscillating electric force pushes electrons up and down, thereby creating oscillating signals in a radio receiver attached to the antenna.

Cosmic radio waves (radio waves coming from outside the Earth) were discovered serendipitously in 1932 by Karl Jansky, a radio engineer at the Bell Telephone Laboratories in Holmdel, New Jersey. Fresh out of college, Jansky had been assigned the task of identifying the noise that plagued telephone calls to Europe. In those days, telephone calls crossed the Atlantic by radio transmission, so Jansky constructed a special radio antenna, made of a long array of metal pipes, to search for sources of radio static (Figure 9.1a). Most of the static, he soon discovered, came from thunderstorms, but when the storms were gone, there remained a faint, hissing static. By 1935 he had identified the source of the hiss; it was coming, mostly, from the central regions of our Milky Way galaxy. When the central regions were overhead, the hiss was strong; when they sank below the horizon, the hiss weakened but did not entirely disappear.

This was an amazing discovery. Anyone who had ever thought about cosmic radio waves had expected the Sun to be the brightest source of radio waves in the sky, just as it is the brightest source of light. After all, the Sun is a billion (10^9) times closer to us than most other stars in the Milky Way, so its radio waves ought to be roughly $10^9 \times 10^9 = 10^{18}$ times brighter than those from other stars. Since there are only 10^{12} stars in our galaxy, the Sun should be brighter than all the others put together by a factor of roughly $10^{18}/10^{12} = 10^6$ (a million). How could this argument fail? How could the radio waves from the distant central regions of the Milky Way be so much brighter than those from the nearby Sun?

As amazing as this mystery might be, it is even more amazing, in retrospect, that astronomers paid almost no attention to the mystery. In fact, despite extensive publicity by the Bell Telephone Company, only two astronomers seem to have taken any interest at all in Jansky's discovery. It was doomed to near oblivion by the same astronomical conservatism that Chandrasekhar was encountering with his claims that no white dwarf can be heavier than 1.4 Suns (Chapter 4).

The two exceptions to this general lack of interest were a graduate student, Jesse Greenstein, and a lecturer, Fred Whipple, in Harvard University's astronomy department. Greenstein and Whipple, pondering